SQL Challenge: Big Companies

Problem Statement

*An organization maintains employment data in three tables: **EMPLOYEE**, **COMPANY**, and **SALARY**. Write a query to print the names of companies where the average employee salary is greater than 40,000. Each row in the output must contain the name of a company in the **COMPANY.NAME** format.*

Schema

Table: EMPLOYEE

Column	Type	Description
ID	Integer	Employee ID (Primary Key) [1, 1000]
NAME	String	Employee name (1-100 characters)

Table: COMPANY

Column	Type	Description
ID	Integer	Company ID (Primary Key) [1, 1000]
NAME	String	Company name (1-100 characters)

Table: SALARY

Column	Type	Description
EMPLOYEE_ID	Integer	Employee ID [1, 1000]
COMPANY_ID	Integer	Company ID [1, 1000]
SALARY	Integer	Salary [10000, 100000]

SQL Solution

To solve this, we need to join the **COMPANY** table with the **SALARY** table, group by the company name, and filter the groups using the `HAVING` clause for the average salary condition.

```
SELECT
    C.NAME
FROM
    COMPANY C
JOIN
    SALARY S ON C.ID = S.COMPANY_ID
GROUP BY
    C.NAME
HAVING
    AVG(S.SALARY) > 40000;
```

Explanation

- **JOIN:** We link the `COMPANY` and `SALARY` tables using the `COMPANY_ID` field.
- **GROUP BY:** We group the results by `COMPANY.NAME` to calculate statistics for each specific company.
- **AVG():** This aggregate function calculates the mean salary of all employees within that company group.
- **HAVING:** Unlike `WHERE`, `HAVING` is used to filter based on aggregate results (in this case, where the average is $> 40{,}000$).